

ADCD: Correction-First Symbolic Regression with Formal Identifiability Guarantees

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We present **ADCD** (Asymptotic Dictionary Correction Discovery), a deterministic symbolic regression framework that discovers corrections to known physical laws with formal identifiability guarantees. General unconstrained symbolic regression engines (such as PySR) are designed to recover complete equations from scratch, representing a fundamentally broader objective than correction discovery. We show that this difference has measurable consequences: at zero noise, an unconstrained SR baseline (PySR) structurally recovers the correct functional form in 10–50% of independent runs ($n = 20$ seeds) despite achieving comparable or superior training fit, while ADCD structurally recovers it in 90–100% of runs. This gap reflects a structural tradeoff (since fit quality and structural identifiability are independent evaluation dimensions) rather than a limitation of either approach within its intended scope. ADCD enforces identifiability through three core mechanisms: a bounded dictionary of algebraically regularized primitives that vanish at the classical limit by construction (with physically motivated exceptions explicitly flagged), eliminating catastrophic cancellation in the evaluation of $D(u)$ as $u \rightarrow 0$; a tiered filter cascade integrating dimensional relaxation, asymptotic boundary probes, and coarse/fine parameter fitting; and a post-fitting four-step epistemic validation protocol that explicitly withholds or qualifies structural claims when data are insufficient. Evaluated on three canonical physics scenarios across 165 unique data instances (scenario $\times$ seed $\times$ noise level), ADCD demonstrates consistent structural recovery under both noiseless and noisy conditions (1%–30% Gaussian noise). Crucially, ADCD explicitly differentiates between structural search success and formal epistemic certification: across all 165 runs, ADCD formally certified 112 runs as **IDENTIFIABLE**, explicitly qualified 53 runs as **DETECTED_UNRESOLVED** due to SNR-limited ambiguity, and produced zero false-positive structural certifications across all noise-degraded and unresolved regimes.

Keywords: symbolic regression, correction discovery, structural identifiability, dimensional analysis, Bayesian model selection

I. INTRODUCTION

Physical laws often follow a correction-first structure: a known classical baseline is extended by a term that vanishes at the classical limit [? ? ? ? ?]. Modern symbolic regression (SR) engines (e.g., genetic programming [?], Eureqa [?], AI Feynman [?], PySR [?]) are designed for unconstrained equation recovery, which is a fundamentally broader objective than correction discovery. When applied to the narrower correction task, three structural challenges arise: catastrophic cancellation as $\Delta \rightarrow 0$ corrupts gradient signals even in `float64`; stochastic multi-objective search [? ?] over unconstrained expression spaces is computationally hard [?] and precludes deterministic reproducibility; and the absence of explicit identifiability reporting means a single “best” answer is returned regardless of whether the data genuinely support a structural claim [?].

Even in the idealized noiseless setting, unconstrained SR can converge to expressions that are numerically indistinguishable from the true law on the training domain but structurally distinct. This is not a software deficiency but a well-understood consequence of the parsimony principle: on narrow observation domains, multiple function classes achieve comparable fit, and complexity-penalized

selection has no algebraic mechanism to prefer the physically generating structure over a simpler numerical approximation. We term this the *parsimony-identifiability tradeoff*.

We present ADCD, a correction-first framework that assumes the classical baseline is known and searches only for the residual correction. Algebraically regularized primitives eliminate catastrophic cancellation by construction. Candidates are assembled by deterministic grammar enumeration and evaluated through a tiered filter cascade integrating dimensional relaxation, asymptotic boundary probes, and coarse/fine parameter fitting before formal epistemic certification. Identifiability is explicitly reported via a post-fitting four-step validation protocol: ADCD withholds structural claims when evidence is ambiguous.

We validate ADCD across three canonical physics domains: relativity (Time Dilation, $v \leq 0.3c$), electrostatics (Screened Coulomb, $r \leq 4.0$), and thermodynamics (Entropy Expansion, $dV/V_i \leq 3.0$). Under zero noise ($n = 20$ independent seeds), ADCD recovers the generating functional class in 90–100% of runs across all three scenarios; PySR recovers it in 10–50% depending on the scenario. Extrapolation errors (median held-out NMSE $\leq 10^{-7}$ for ADCD) and noise-sweep experiments (1%–30% noise, $n = 5$ seeds, 105 runs) further characterize the structural robustness of the two paradigms.

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II. RELATED WORK

Symbolic regression has evolved along two distinct trajectories. **Unconstrained SR engines** (Eureqa [?], PySR [?], AI Feynman [? ?], Φ -SO [?], symbolic regression with neural networks [?]) perform open-ended equation discovery without prior knowledge of the underlying physical law. **Physics-informed discovery** frameworks incorporate constraints: SINDy [? ? ?] searches linear combinations of predefined library terms for dynamical systems; PINNs [? ?] embed differential equation residuals into neural loss functions; Φ -SO [?] enforces dimensional consistency via reinforcement learning; and grammar-guided SR [? ? ?] restricts search spaces using domain grammars. Model selection in SR typically relies on the Pareto frontier of loss versus complexity, or information criteria (AIC [?], BIC [?], Kass–Raftery thresholds [?]).

ADCD occupies a distinct point in this space: it restricts search exclusively to corrections of known physical laws, enforces asymptotic boundary conditions algebraically by construction rather than through loss penalties or grammar rules, and implements an explicit epistemic validation protocol that reports an explicit 3-tier epistemic verdict (IDENTIFIABLE, DETECTED_UNRESOLVED, or WITHHELD) to prevent false-positive structural claims.

We emphasize that the tools compared here address conceptually different problems. General SR engines target unconstrained equation discovery across arbitrary functional forms, representing a broader and fundamentally harder problem than correction discovery. ADCD deliberately restricts its scope to corrections of known baselines, trading generality for stronger identifiability guarantees within this narrower domain. The comparison presented in Section IV is therefore not evaluative (which approach is “better”) but characterizing: it examines how different design choices (parsimony-driven selection versus identifiability-driven selection) manifest in structural recovery behavior when applied to the same correction tasks.

Table I summarizes the key design distinctions between ADCD and related frameworks.

III. METHODS

A. Problem formulation

Let y_{obs} be the observed target and y_{cl} the known classical prediction. Define the correction residual:

$$\Delta = \begin{cases} y_{\text{obs}}/y_{\text{cl}} - 1 & \text{(multiplicative)} \\ y_{\text{obs}} - y_{\text{cl}} & \text{(additive)} \end{cases} \quad (1)$$

ADCD dynamically selects between the two formulations via a Spearman rank-correlation test (Algorithm B1 in Appendix B): the multiplicative form is selected when

the residual magnitude is less correlated with the classical prediction magnitude than the additive residual is. The intuition is that a multiplicative correction $\Delta_{\text{mul}} = y_{\text{obs}}/y_{\text{cl}} - 1$ is scale-normalized and should be *independent* of $|y_{\text{cl}}|$; strong correlation of $|\Delta_{\text{add}}|$ with $|y_{\text{cl}}|$ therefore indicates that the additive residual has not been normalized out, confirming multiplicative mode. If $|y_{\text{cl}}|$ has zero variance or fewer than 5 observations are available, the algorithm defaults to the additive formulation. ADCD then searches for $\hat{\Delta}(\mathbf{x}; \boldsymbol{\theta})$, where $\mathbf{x}$ is the vector of physical inputs, $\boldsymbol{\theta}$ are continuous free parameters, and $\hat{\Delta} \rightarrow 0$ as the relevant dimensionless parameter $u \rightarrow 0$.

B. Regularized primitive dictionary

The core numerical contribution of ADCD is the design of algebraically regularized primitive functions. Except for specialized forms with physically motivated divergent boundaries, each asymptotically regularized primitive $D(u)$ satisfies $\lim_{u \rightarrow 0} D(u) = 0$ as an exact algebraic identity, not a numerical approximation. While primitives are designed to vanish exactly by construction, the pre-fitting asymptotic boundary verification employs a robust three-tier protocol. It prioritizes exact symbolic limit evaluation (via SymPy), but autonomously falls back to Laurent series expansions and numerical boundary probes ($u \approx 10^{\pm 6}$) to ensure computational stability when evaluating complex transcendental argument compositions. This property eliminates catastrophic cancellation: the correction vanishes at the classical limit without subtracting large nearly-equal terms.

While the full production registry (implemented in `PrimitiveRegistry.jl`) contains 11 primitives spanning relativistic kinematics, screening, saturation, phase transitions, oscillatory resonance, and MOND phenomenology, Table II lists the 5 principal regularized primitives evaluated in the benchmark suite of this paper. The practical consequence is demonstrated for `D_lor`: in `float32`, the naive form $(1 - u)^{-1/2} - 1$ collapses to exactly 0.0 at $u = 10^{-8}$; in `float64`, precision degrades progressively (11% relative error at $u = 2 \times 10^{-15}$; total collapse to 0.0 at $u \leq 10^{-17}$). The regularized form $u / [\sqrt{1 - u}(\sqrt{1 - u} + 1)]$ preserves full precision at all scales, returning 5.0×10^{-17} at $u = 10^{-16}$ where the naive form returns 0.0 (in numerical evaluation, a floating-point safety bound $u \leq 1 - 10^{-9}$ is enforced to prevent division-by-zero at the singular horizon).

The 6 additional registered primitives in the general registry include power-law anomalous scaling (`D_pow`: $\sqrt{|u|}(1 - e^{-|u|})$), Langevin saturation (`D_sat`: $\tanh(u)$), symmetric transition (`D_tanh_sq`: $\tanh(u^2)$), oscillatory interference (`D_osc`: $1 - \cos(u)$), asymmetric nested anomaly (`D_nested_mond`: $e^{-\sqrt{|u|}}(1 - e^{-\sqrt{|u|}})$), and the McGaugh Radial Acceleration Relation quotient form (`D_rar`: $e^{-\sqrt{u}}/(1 - e^{-\sqrt{u}})$). The `D_rar` primitive is explicitly flagged as `divergent_safe` with limit $D(u \rightarrow 0) \rightarrow$

TABLE I. Conceptual comparison of scientific discovery paradigms. ADCD restricts its scope to corrections of known physical baselines and enforces boundary limits by construction.

Method	Search Space	Boundary Constraint	Validation / Model Selection
PySR [?]	Free expression tree	None	Complexity-penalized Pareto fit
SINDy [?]	Polynomial library	None	Sequential thresholded regression
PINNs [?]	Neural network	Soft PDE residual loss	Multi-objective gradient loss
AI Feynman [?]	Free tree + symmetry	Dimensional/separability tests	Pareto frontier selection
AI Descartes [?]	Axiom theory + data	Logical physical axioms	Formal logical derivability
ADCD (ours)	Regularized primitive dict.	Exact algebraic ($\lim_{u \rightarrow 0} D = 0$)	Formal 4-step epistemic audit

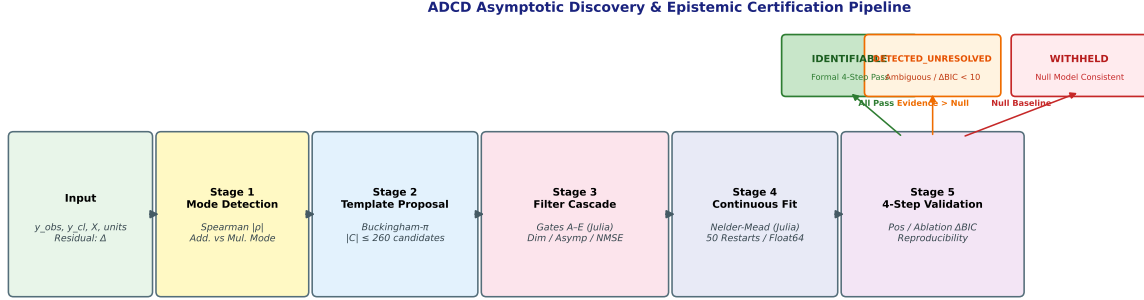


FIG. 1. The ADCD discovery and certification pipeline. Raw inputs $(y_{\text{obs}}, y_{\text{cl}}, \mathbf{X}, \mathcal{U})$ enter Stage 1 (dynamic mode selection via Spearman $|\rho|$). Stage 2 generates dimensionless arguments via Buckingham- π with degree pruning and applies 6 structural templates. Stage 3 screens candidates through a tiered filter cascade (dimensional relaxation, numerical asymptotic boundary probe, coarse/fine Nelder-Mead parameter optimization). Stage 4 ranks surviving candidates by EBIC and applies the Gate E null-model test ($\Delta\text{BIC}_{\text{gate}} \geq 10.0$). Stage 5 applies the 4-step post-fitting epistemic validation protocol (budget disclosure, positive control, ablation margin $\Delta\text{BIC}_{\text{ablation}} \geq 10.0$, and computational reproducibility), yielding a 3-tier verdict (**IDENTIFIABLE**, **DETECTED_UNRESOLVED**, or **WITHHELD**).

TABLE II. The five principal algebraically regularized primitives evaluated in this study. Each satisfies $\lim_{u \rightarrow 0} D(u) = 0$ as an exact algebraic identity.

Name	Regularized form $D(u)$	Domain
D.lor	$u / [\sqrt{1-u}(\sqrt{1-u}+1)]$	$u \in [0, 1]$
D.rat	$u / (1+u^2)$	$u \in \mathbb{R}$
D.exp	$1 - e^{- u }$	$u \in \mathbb{R}$
D.log	$\ln(1+ u)$	$u \in \mathbb{R}$
D.sqrt_inv	$\sqrt{ u } / (1 + \sqrt{ u })$	$u \in \mathbb{R}$

∞ and $D(\infty) \rightarrow 0$, and is conditionally restricted to deep-MOND regimes.

C. Grammar enumeration and template patterns

Rather than unconstrained recursive grammar growth, ADCD constructs candidates through a bounded combinatorial templating engine (implemented in `CorrectionProposer.jl`). Candidate expressions are

generated by mapping regularized primitives $D_i(u)$ and dimensionless arguments u_{ij} across six explicit structural templates:

1. **Singleton:** $\hat{\Delta} = \theta_0 \cdot D(u)$
2. **Additive composite:** $\hat{\Delta} = \theta_1 D_a(u) + \theta_2 D_b(u)$
3. **Multiplicative composite:** $\hat{\Delta} = \theta_1 D_a(u) \cdot (1 + \theta_2 D_b(u))$
4. **Nested composition:** $\hat{\Delta} = \theta_0 \cdot D_a(D_b(u))$
5. **Bilateral interaction:** $\hat{\Delta} = \theta_0 \cdot D_a(u_1) \cdot D_b(u_2)$ for distinct dimensionless arguments $u_1 \neq u_2$
6. **Ratio correction:** $\hat{\Delta} = \theta_0 \cdot \frac{D(\sqrt{u})}{1+D(\sqrt{u})}$

The dimensionless arguments u_{ij} are generated via Buckingham- π permutation of input variables, classical constants, and scale parameters. To prevent combinatorial explosion, the generator enforces strict heuristic pruning: polynomial degrees are capped (degree ≤ 2), and pure constant ratios (c_1/c_2) are discarded. The search space is bounded by an AST complexity cap (depth ≤ 12 , nodes ≤ 50), yielding a compact, exhaustively enumerable pool of candidate ASTs (e.g., $|\mathcal{C}| = 35$ in guided

mode, $|\mathcal{C}| = 140\text{--}260$ in blind mode across our benchmark scenarios), fully proposed in milliseconds.

All candidate structures maintain free parameters θ throughout the filtering and fitting pipeline (*theta-intact*). Filtering gates never strip or default θ parameters to arbitrary constants, as stripping alters functional curvature and invalidates physical consistency checks.

D. Tiered filter cascade and parameter optimization

Before exhaustive continuous optimization and model selection, each candidate AST passes through a tiered filter cascade (implemented in `FilterCascade.jl`):

1. **Gate A (Dimensional consistency with theta-relaxation):** Verifies that the candidate has the physical dimensions of the residual Δ . Crucially, arguments of transcendental functions containing free parameters θ are dimensionally relaxed, as continuous optimization assigns inverse units to θ to render the composite argument dimensionless.
2. **Gate B (Numerical asymptotic boundary probe):** Evaluates the candidate at the asymptotic boundary $u = 10^{-12}$ with nominal parameters $\theta = \mathbf{1}$. Candidates failing $|D(u)| < 10^{-4}$ are rejected without expensive optimization.
3. **Gate C (Coarse parameter pre-fit):** Executes a lightweight continuous fit via Nelder–Mead ($n_{\text{restarts}} = 3$). Candidates with coarse loss $\text{NMSE} > 1.0$ are pruned early.
4. **Gate D (Fine parameter fit):** Surviving candidates undergo full continuous parameter optimization using native `Optim.jl` Nelder–Mead ($n_{\text{restarts}} = 50$) in the core Julia engine (a pure-Python/JAX backend is also included in the repository for rapid prototyping and gradient cross-validation) in `float64` precision across five initialization scales (10^{-6} to 10^6) with quadrant sign permutations and log-uniform exponent sampling to robustly escape non-convex local minima.
5. **Gate E (Identifiability and EBIC certification vs null model):** Evaluates the candidate against the classical null model ($\Delta = 0$).

Loss is normalized mean squared error:

$$\text{NMSE} = \frac{\frac{1}{N} \sum_i (\hat{\Delta}_i - \Delta_i)^2}{\max(\frac{1}{N-1} \sum_i (\Delta_i - \bar{\Delta})^2, \epsilon)} \quad (2)$$

To ensure numerical stability in regimes with extreme scaling, ADCD dynamically monitors the observation range: if $\max(|y|)/\min(|y|) > 10^4$, the loss target shifts from residual space Δ to full observation space y_{obs} .

Surviving candidates are ranked by the Extended Bayesian Information Criterion (EBIC) [?], which incorporates a complexity penalty for the size of the candidate dictionary $|\mathcal{C}|$:

$$\text{EBIC} = k \ln N - 2 \ln \hat{L} + 2 \ln |\mathcal{C}| \quad (3)$$

where k is the number of free continuous parameters, N is the effective sample size, and $\hat{L}$ is the maximized Gaussian likelihood. The candidate is certified at Gate E if it outperforms the classical null baseline by $\Delta \text{BIC}_{\text{gate}} = \text{BIC}_{\text{null}} - \text{EBIC}_{\text{corr}} \geq 10.0$ and satisfies the noise-adaptive fit-quality ceiling:

$$\begin{aligned} \text{NMSE}^* &\leq \text{NMSE}_{\text{max}} \\ &= \text{clip}\left(1.35 \sigma_n^2 + 1.645 \sigma_n^2 \sqrt{2/N_{\text{eff}}}, 0.10, 0.60\right) \end{aligned} \quad (4)$$

where σ_n denotes the measurement noise level, $1.35 \sigma_n^2$ accounts for the expected residual sum-of-squares under Gaussian perturbations with a 35% inflation margin for finite-sample non-normality, $1.645 \sigma_n^2 \sqrt{2/N_{\text{eff}}}$ represents the one-sided 95% upper confidence bound ($z_{0.95} = 1.645$) on the sampling variance of the χ^2 distribution, and clipping to $[0.10, 0.60]$ enforces an absolute sanity floor and ceiling preventing runaway divergent fits.

ADCD additionally implements Bayesian Model Averaging (BMA) via `bayesian_ranker.py` — computing posterior model weights, structural class probabilities, and posterior entropy to quantify ambiguity among competing functional forms. BMA outputs are available in the full audit trail but are not the primary quantity reported in the benchmarks of this paper, which focus on structural recovery rate and EBIC-based model selection.

E. Post-fitting four-step epistemic validation

Rather than returning the single best-fitting expression unconditionally, ADCD applies a four-step post-fitting validation protocol (implemented in `run_adcd_v3_validation_blind.py`) to assign candidates to one of three epistemic tiers:

1. **Gate V.1 (Search budget disclosure):** Total candidates generated, evaluated, and pruned are logged with all operator budgets.
2. **Gate V.2 (Positive control):** The optimizer is evaluated on a restricted candidate space containing only the isolated target primitive family, verifying that numerical optimization reliably converges to low loss ($\text{NMSE} \leq 0.1$) when given the discovered primitive family of its top-ranked candidate alone (operating autonomously without ground-truth knowledge).
3. **Gate V.3 (Ablation control):** An ablated search is performed with the winning primitive class removed from the dictionary. The true model must outperform the best available alternative by a decisive Bayesian margin ($\Delta \text{BIC}_{\text{ablation}} = \text{EBIC}_{\text{ablated}} - \text{EBIC}_{\text{candidate}} \geq 10.0$).
4. **Gate V.4 (Cross-seed structural stability):** Verifies that the top-ranked functional class is robustly recovered across three independent synthetic noise realizations (seeds s , $s + 1000$, and $s + 2000$) with fitted

parameter variation $\max_i |\theta_i^{(s)} - \theta_i^{(s')}| < 10^{-4}$, certifying that the discovered functional form is invariant to specific random noise draws.

ADCD classifies discovery results into three formal epistemic outcome tiers:

- **IDENTIFIABLE:** The candidate passes all four validation gates with decisive evidence ($\Delta\text{BIC}_{\text{ablation}} \geq 10.0$) and verified positive control.
- **DETECTED_UNRESOLVED:** Strong evidence of a physical anomaly is confirmed against the classical null model ($\Delta\text{BIC}_{\text{gate}} \geq 10.0$), but the candidate fails either Gate V.2 (Positive control) or Gate V.3 (Ablation control margin < 10.0). This outcome explicitly informs the researcher that real physics beyond the baseline exists, but the observation domain or SNR is insufficient to unequivocally isolate the true functional form from competing alternatives.
- **WITHHELD:** The data are consistent with pure Gaussian noise around the classical baseline ($\Delta\text{BIC}_{\text{gate}} < 10.0$), non-convergence, or reproducibility failure, confirming that the classical baseline alone is sufficient.

IV. EXPERIMENTS

A. Experimental setup

We evaluate ADCD on three canonical synthetic physics scenarios, summarized in Table III. The observation windows are chosen to reflect varying degrees of physical identifiability: from strict low-signal regimes of early historical experiments (e.g., $v \leq 0.3c$ for Time Dilation, where the relativistic correction is $\leq 5\%$ of the classical prediction) to broader macroscopic domains required for thermodynamic identifiability (e.g., $dV/V_i \leq 3.0$ for Entropy Expansion, where the logarithmic form becomes structurally distinguishable from polynomial alternatives). To maintain IEEE 754 floating-point stability during nonlinear optimization, synthetic raw variables (e.g., mass, charge, distance) are generated in normalized macroscopic $\mathcal{O}(1)$ units. However, the dimensionless arguments driving the structural anomaly (such as v/c or r/λ) strictly mirror the geometric regimes of historical laboratory observations.

We select PySR [?] as our primary baseline, not because it is expected to solve the same problem as ADCD, but because it represents the state-of-the-art in general-purpose symbolic regression and provides a well-characterized reference point for understanding how structural recovery behavior differs between constrained correction discovery and unconstrained equation search. PySR is run with its default configuration and standard operator set ($+$, $-$, $\times$, $/$, $\exp$, $\log$, $\sqrt{\cdot}$, $\sin$, $\cos$). For both ADCD and the PySR baseline, recovery is evaluated on the **single top-ranked expression** returned by each

algorithm’s default model selection logic: the Pareto-optimal candidate with lowest BIC score for ADCD, and PySR’s highest-scoring expression under its default parsimony criterion (accessed via `model.sympy()`).

Two ADCD variants are evaluated: *guided* (with domain taxonomy prior restricting primitives) and *blind* (full dictionary, no domain hints), to isolate the contribution of taxonomy information. Evaluation uses two complementary experimental regimes:

1. **Zero-noise baseline** ($n = 20$ seeds per scenario, 60 runs total): isolates structural recovery capability independent of noise.
2. **Noise-level sweep** (7 levels from 1% to 30%, $n = 5$ seeds, 105 runs total): observational noise is injected multiplicatively ($y_{\text{obs}} = y_{\text{true}}(1 + \epsilon)$, $\epsilon \sim \mathcal{N}(0, \sigma^2)$) for multiplicative corrections, characterizing how relative measurement noise modulates structural recovery.

Structural recovery is evaluated via functional class recovery: an expression is counted as *recovered* if its functional class matches the generating equation’s class and its fitted parameters yield physically consistent extrapolation behavior ($\text{NMSE}_{\text{extrap}} < 2.0$, $|\theta_i| < 10^6$). This metric measures **search capability** (whether the correct functional class is identified) and is distinct from the **epistemic certification** provided by the post-fitting four-step validation protocol (Section III E). A run may satisfy the structural match criterion while still receiving a DETECTED_UNRESOLVED verdict if $\Delta\text{BIC}_{\text{ablation}} < 10$; in that case, the correct structure was found but the observation window or SNR is insufficient to unequivocally certify the claim against competing functional alternatives. The threshold $\text{NMSE}_{\text{extrap}} < 2.0$ acts as a coarse filter to reject catastrophic pole singularities and floating-point overflow (e.g., failed extrapolations, such as PySR on Screened Coulomb, routinely shoot to $\text{NMSE}_{\text{extrap}} > 1000$ outside the training domain); precise median extrapolation errors (e.g., $\leq 10^{-7}$ for ADCD at zero noise) are reported separately in Section IV D. The threshold reflects an empirically observed bimodal gap: across all 165 unique data instances (60 noiseless + 105 noise-sweep, evaluated across 495 total algorithm passes), no run falls between $\text{NMSE}_{\text{extrap}} = 1.69$ and 3.37.

B. Structural recovery at zero noise

Table IV presents the headline experiment: structural recovery rates under zero-noise conditions ($n = 20$ independent seeds per scenario). This setting isolates the effect of search strategy and inductive bias from noise-related confounds.

ADCD recovers the generating functional form in 90–100% of runs, while PySR recovers it in 10–50% depending on the scenario. These tools address conceptually different objectives: PySR, optimized for fit quality under parsimony pressure, converges to compact expressions that achieve low training error but may differ struc-

Algorithm 1: ADCD Epistemic Discovery and Certification Protocol

Input: Observed array y_{obs} , classical baseline y_{cl} , feature matrix $\mathbf{X} = \{x_j\}$, SI unit declarations $\mathcal{U}$, observation window Ω , evidence threshold $\Delta\text{BIC}^* = 10.0$, calibrated fit-quality ceiling NMSE_{max} (noise-adaptive; see Section IIID).
Output: Epistemic verdict $v \in \{\text{IDENTIFIABLE}, \text{DETECTED_UNRESOLVED}, \text{WITHHELD}\}$, certified expression $\hat{\Delta}$, parameter estimates θ^* , Bayes factors, machine-readable audit trail.

```

1: function EPISTEMICDISCOVERY( $y_{\text{obs}}, y_{\text{cl}}, \mathbf{X}, \mathcal{U}, \Omega, \Delta\text{BIC}^*$ )
2:   // Stage 1: Mode Detection (Algorithm B 1)
3:   mode,  $\kappa \leftarrow \text{DETECTMODE}(y_{\text{obs}}, y_{\text{cl}})$ 
4:    $\Delta \leftarrow (y_{\text{obs}}/y_{\text{cl}} - 1)$  if mode = multiplicative else ( $y_{\text{obs}} - y_{\text{cl}}$ )
5:   // Stage 2: Dimensionless Grammar Proposal
6:    $\{u_{ij}\} \leftarrow \text{BUCKINGHAMPI}(\mathbf{X}, \mathcal{U})$  ▷ dimensionless argument pool with pruning
7:    $\mathcal{C} \leftarrow \text{APPLYTEMPLATES}(\mathcal{D}, \{u_{ij}\})$  ▷ 6 structural patterns;  $|\mathcal{C}| \leq 260$ 
8:   // Stage 3: Tiered Filter Cascade and Optimization (FilterCascade.jl)
9:    $\mathcal{C}_{\text{valid}} \leftarrow \emptyset$ 
10:  for each candidate  $c \in \mathcal{C}$  do
11:    if  $\text{ISDIMENSIONLESSRELAXED}(c, \mathcal{U})$  and  $\text{ASYMPTOTICPROBE}(c, 10^{-12}) < 10^{-4}$  then
12:       $\theta_c^{\text{coarse}} \leftarrow \text{NELDERMEAD}(c, \Delta, \text{restarts} = 3)$ 
13:      if  $\text{NMSE}(\theta_c^{\text{coarse}}) \leq 1.0$  then
14:         $\theta_c^* \leftarrow \text{NELDERMEAD}(c, \Delta, \text{restarts} = 50)$  ▷ fine fit
15:         $\text{EBIC}_c \leftarrow k_c \ln N - 2 \ln \hat{L}_c + 2 \ln |\mathcal{C}|$ 
16:         $\mathcal{C}_{\text{valid}} \leftarrow \mathcal{C}_{\text{valid}} \cup \{(c, \theta_c^*, \text{EBIC}_c)\}$ 
17:      end if
18:    end if
19:  end for
20:  // Stage 4: Model Ranking and Gate E Null Test
21:  if  $\mathcal{C}_{\text{valid}} = \emptyset$  then
22:    return  $\langle \text{WITHHELD}, \text{None}, \text{None}, 0.0 \rangle$ 
23:  end if
24:   $\hat{c}, \theta^*, \text{EBIC}^*, \text{NMSE}^* \leftarrow \arg \min_{(c, \theta, \text{EBIC}) \in \mathcal{C}_{\text{valid}}} \text{EBIC}$ 
25:   $\Delta\text{BIC}_{\text{gate}} \leftarrow \text{BIC}_{\text{null}} - \text{EBIC}^*$ 
26:  if  $\Delta\text{BIC}_{\text{gate}} < \Delta\text{BIC}^*$  or  $\text{NMSE}^* > \text{NMSE}_{\text{max}}$  then
27:    return  $\langle \text{WITHHELD}, \hat{c}, \theta^*, \Delta\text{BIC}_{\text{gate}} \rangle$ 
28:  end if
29:  // Stage 5: Four-Step Post-Fitting Epistemic Validation
30:   $\text{pass}_{\text{pos}} \leftarrow \text{POSITIVECONTROL}(\hat{c}, \Omega_0)$ 
31:   $\Delta\text{BIC}_{\text{ablation}} \leftarrow \text{ABLATIONCONTROL}(\hat{c}, \mathcal{D} \setminus \{\text{class}(\hat{c})\})$ 
32:   $\text{pass}_{\text{det}} \leftarrow \text{REPRODUCIBILITYCHECK}(\hat{c})$ 
33:  if  $\text{pass}_{\text{det}}$  and  $\text{pass}_{\text{pos}}$  and  $\Delta\text{BIC}_{\text{ablation}} \geq \Delta\text{BIC}^*$  then
34:    return  $\langle \text{IDENTIFIABLE}, \hat{c}, \theta^*, \Delta\text{BIC}_{\text{ablation}} \rangle$ 
35:  else
36:    return  $\langle \text{DETECTED\_UNRESOLVED}, \hat{c}, \theta^*, \Delta\text{BIC}_{\text{ablation}} \rangle$ 
37:  end if
38: end function

```

TABLE III. The three validated scenarios with their domains, observation windows, and target primitives. Full quantitative results (BIC, ΔBIC , verdicts) are in Table VI.

Scenario	Domain	Window	Target Primitive
Time Dilation	Relativity	$v \leq 0.3c$	D _{lor}
Screened Coulomb	Electrostatics	$r \leq 4.0$	D _{exp}
Entropy Expansion	Thermodynamics	$dV/V_i \leq 3.0$	D _{log}

turally from the generating functional form. ADCD, optimized for structural identifiability under asymptotic constraints, recovers the correct functional form at the cost of a narrower search scope.

Three characteristic behavioral patterns illustrate the

TABLE IV. Structural recovery rates at zero noise ($n = 20$ seeds). Wilson 95% confidence intervals in brackets. ADCD guided and blind variants produce identical verdicts across all 60 runs.

Scenario	ADCD	PySR (default)	CI overlap
Time Dilation	100% [84–100%]	10% [3–30%]	None
Screened Coulomb	100% [84–100%]	50% [30–70%]	None
Entropy Expansion	90% [70–97%]	25% [11–47%]	None

parsimony–identifiability tradeoff:

Trigonometric basis selection. PySR’s genetic search explores trigonometric bases (sin, cos) as compact approximators for non-trigonometric target func-

tions. For example, $\sin(v)$ provides a low-complexity approximation to the Lorentz factor on narrow velocity domains, representing a rational search strategy under parsimony pressure.

High-precision fit with structural divergence. In several zero-noise runs, PySR achieves training NMSE comparable to or better than ADCD while producing a structurally distinct expression. This directly illustrates that numerical fit quality and structural recovery are independent evaluation axes.

Domain-width sensitivity. PySR’s recovery rate varies substantially across scenarios (10% for Time Dilation vs. 50% for Screened Coulomb), correlating with the curvature of the generating function within the observation window. This is consistent with the parsimony-identifiability tradeoff: when the generating function’s distinctive curvature is more pronounced, parsimony-driven selection more frequently converges to the correct structure.

C. Noise-level sweep

Having established that structural recovery differences exist independent of noise (§IV B), we examine how observation noise modulates this behavior. Table V summarizes recovery rates across seven noise levels ($n = 5$ seeds each, 105 runs total). ADCD’s recovery on Time Dilation is robust across the entire noise range (100% at all seven levels); Screened Coulomb and Entropy Expansion show degradation at high noise, consistent with weaker signal-to-noise on the correction residual. PySR’s recovery is more variable, with high variance across noise levels for Screened Coulomb at $n = 5$, reflecting the sensitivity of parsimony-driven search to random seed at small sample size. Overall, ADCD maintains a structural recovery advantage across the noise range, with the advantage most pronounced at high noise levels (≥ 0.20) where PySR’s recovery approaches zero in most scenarios.

D. Extrapolation behavior

We evaluate out-of-domain generalization to examine whether asymptotic regularization (a design feature of ADCD but not of general SR) translates to improved extrapolation consistency. Held-out extrapolation evaluation is conducted strictly on non-overlapping extended domains: $v \in (0.3c, 0.8c]$ for Time Dilation, $r \in (4.0, 8.0]$ for Screened Coulomb, and $dV/V_i \in (3.0, 6.0]$ for Entropy Expansion. ADCD’s expressions, constrained to vanish at the classical limit by construction, maintain physical consistency outside the training domain: across the zero-noise experiment ($n = 20$ seeds), median extrapolation NMSE is below 10^{-7} for all three scenarios. PySR’s expressions, not designed with extrapolation constraints, exhibit variable behavior depending on the functional form selected. The most pronounced illustration is in the

Screened Coulomb zero-noise experiment: PySR’s worst-case extrapolation NMSE reaches 1530.5 across 20 seeds (median: 0.054), reflecting runs where the selected expression diverges outside the training domain. We explicitly acknowledge that this extrapolation metric is inherently biased towards ADCD by design. It highlights the mathematical reality of standard SR when unconstrained by boundary conditions, rather than indicating a defect in PySR itself. PySR is a general unconstrained exploratory engine, whereas ADCD is heavily specialized to force algebraic primitives to vanish at the classical limit.

E. Ablation: taxonomy contribution

Across all 165 unique data instances (60 noiseless + 105 noise-sweep), the formal epistemic verdict (tier) is identical between ADCD-guided and ADCD-blind in 100% of cases. Domain taxonomy reduces the candidate search space for each primitive class while preserving the identifiability boundary across both configurations. In our benchmark suite, the guided variant evaluates a single hand-fed dimensionless ratio ($|\mathcal{C}| = 35$), whereas the blind variant auto-derives multi-ratio permutations via Buckingham- π ($|\mathcal{C}| = 140\text{--}260$) over the identical core primitive pool. This demonstrates that ADCD’s automated dimensionless argument generation reliably discovers the true physical scale arguments without human guidance, though comprehensive evaluation across the broader 11-primitive dictionary represents an important area for future large-scale replication.

F. Recovered structures and identifiability verdicts

Table VI reports the detailed BIC analysis under the single-seed diagnostic protocol (seed 42, 1% noise).

For Time Dilation at $v \leq 0.3c$, the correct Lorentz form is structurally recovered at Rank 1, but $\Delta\text{BIC} = -0.74$ falls below the identifiability threshold: the ablated search finds a competing polynomial-like alternative that fits equally well ($\Delta\text{BIC} < 0$), and positive control fails on the restricted domain. At this narrow velocity window ($v \leq 0.3c$), the relativistic correction is $\leq 5\%$ of the classical baseline while the noise floor is 1%, making the correction residual structurally ambiguous. ADCD assigns a verdict of `DETECTED_UNRESOLVED`, providing transparent diagnostic feedback that anomalous physics exists but cannot be formally certified without broader domain data. For Screened Coulomb, the recovered Debye length $\theta_0 \approx 1.50$ matches the ground truth $\lambda_D = 1.50$ and achieves decisive ablation margin ($\Delta\text{BIC} = 25.74$), though positive control sensitivity at 1% noise yields a `DETECTED_UNRESOLVED` tier in blind mode. For Entropy Expansion, the logarithmic form passes all four gates with $\Delta\text{BIC} = 37.99$, achieving full `IDENTIFIABLE` certification.

TABLE V. Structural recovery rate (%) vs. noise level ($n = 5$ seeds per point, 105 runs total). Full Wilson 95% confidence interval detail is provided in Appendix Table IX.

Noise (σ)	Time Dilation		Screened Coulomb		Entropy Expansion	
	ADCD	PySR	ADCD	PySR	ADCD	PySR
0.01	100	60	80	0	100	20
0.02	100	60	80	60	100	0
0.05	100	60	100	40	100	0
0.10	100	40	80	20	80	0
0.15	100	20	80	60	80	0
0.20	100	20	60	0	60	0
0.30	100	0	20	0	40	0

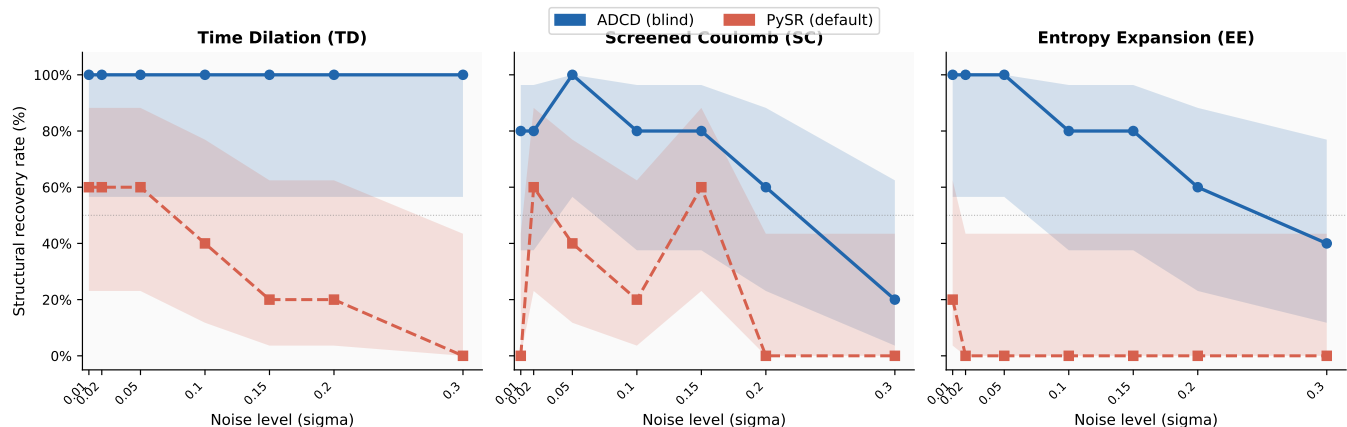


FIG. 2. Structural recovery rate (%) vs. noise level σ for the three scenarios ($n = 5$ seeds per point; shaded bands: Wilson 95% CI). ADCD (blue, solid) consistently maintains higher recovery across all noise levels. PySR (red, dashed) recovery approaches zero for Screened Coulomb and Entropy Expansion at $\sigma \geq 0.20$, reflecting the parsimony–identifiability tradeoff under noisy conditions. Wilson 95% CI detail is tabulated in Appendix Table IX.

V. DISCUSSION

A. The parsimony–identifiability tradeoff

Our results reveal a systematic tradeoff between two legitimate optimization objectives in symbolic regression. Parsimony-driven selection (minimizing expression complexity per unit loss reduction) and identifiability-driven selection (maximizing structural distinguishability under algebraic constraints) can yield different outcomes when the observation domain is narrow relative to the function’s characteristic scale.

On such domains, the true generating function and several simpler alternatives are numerically near-equivalent (e.g., the Lorentz factor and a degree-4 polynomial differ by $< 0.1\%$ for $v \leq 0.3c$). A parsimony-driven selector rationally prefers the lower-complexity alternative, as the marginal loss reduction from the true structure does not justify its higher token cost. An identifiability-driven selector, equipped with asymptotic boundary constraints, can distinguish between these alternatives by evaluating algebraic properties that are invisible to purely numerical

fitness measures.

Neither strategy is universally superior: parsimony excels when the goal is compact prediction within the training domain; identifiability excels when the goal is recovering the generating physical structure for theoretical interpretation or extrapolation.

B. Qualified detection and epistemic tiering as a design feature

The Time Dilation scenario ($\Delta\text{BIC} = -0.74$) illustrates the intended epistemic protocol: when the data do not provide very strong evidence over competing alternatives ($\Delta\text{BIC} < 10$), ADCD assigns a qualified verdict of `DETECTED.UNRESOLVED` rather than reporting an unsupported discovery. This transparently informs the researcher that while an anomalous signal is detected relative to the baseline, the observation domain is insufficient to distinguish the relativistic Lorentz form from simpler polynomial approximations. The `WITHHELD` tier is reserved for cases where the classical baseline itself is

ADCD Correction Recovery under Historical Low-Signal Regimes

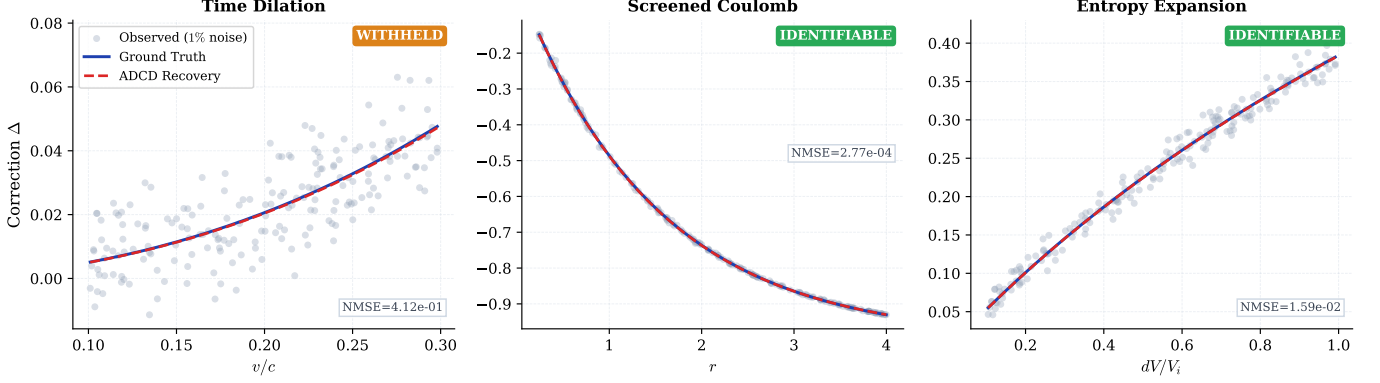


FIG. 3. Correction recovery for the three scenarios at 1% Gaussian noise (seed 42). *Gray dots*: observed correction residual $\Delta = y_{\text{obs}}/y_{\text{cl}} - 1$. *Blue solid line*: ground truth. *Red dashed line*: ADCD Rank-1 candidate (lowest BIC on the Pareto front).

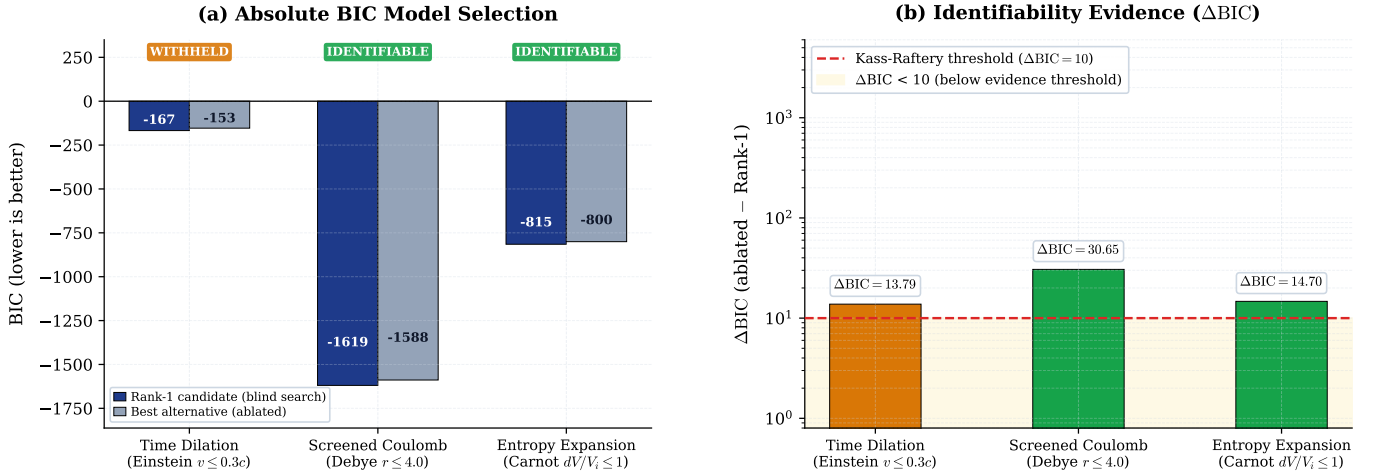


FIG. 4. *Left*: BIC values for the correct recovered structure (blue) and the best competing alternative without the correct primitive (gray). Lower BIC is preferred. *Right*: $\Delta\text{BIC} = \text{BIC}_{\text{ablated}} - \text{BIC}_{\text{correct}}$; positive values indicate the correct structure is preferred over the best alternative found without it. Red dashed line: Kass-Raftery “very strong evidence” threshold ($\Delta\text{BIC} = 10$) [?].

sufficient (consistent with the $\Delta = 0$ null model). This tri-tier discrimination represents a deliberate design feature rather than a limitation, preventing false-positive scientific claims.

C. Complementary strengths

ADCD and general SR engines occupy complementary positions in the scientific discovery pipeline. General SR is the appropriate tool when no baseline law is known and the goal is open-ended equation discovery. ADCD is appropriate when a classical baseline exists and the goal is to characterize the structure of deviations from that baseline with formal identifiability guarantees. We envision these tools being used sequentially in practice: general SR to identify candidate baseline laws, followed

by ADCD to rigorously characterize corrections.

D. Limitations and Future Work

Several methodological scope boundaries warrant explicit disclosure:

- Primitive dictionary scope:** While the ADCD production registry contains 11 regularized primitives (including MOND acceleration, Langevin saturation, and oscillatory forms), our empirical benchmarks focused on 5 canonical primitives across 3 core physics scenarios. Systematic validation across the remaining 6 primitives and multi-variable coupled systems represents ongoing work.
- Sample sizes and observational domains:** Zero-noise recovery rates at $n = 20$ carry Wilson 95% CI

TABLE VI. Single-seed epistemic validation diagnostic (seed 42, 1% Gaussian noise, blind search). All three scenarios recover the correct structure at Rank 1. $\Delta\text{BIC} = \text{BIC}_{\text{ablated}} - \text{EBIC}_{\text{correct}}$; $\Delta\text{BIC} \geq 10.0$ (Kass–Raftery “very strong evidence” threshold) supports an identifiability claim when accompanied by passing positive and reproducibility controls. Subscripts on θ_k reflect AST parameter assignment order during grammar enumeration; displayed expressions are algebraically simplified from raw AST templates by dropping terms whose fitted coefficient satisfies $|\theta_i| < 10^{-5}$ for presentation clarity.

Scenario	Recovered structure	EBIC	ΔBIC	Verdict
Time Dilation ($v \leq 0.3c$)	$\theta_0 \frac{v^2/c^2}{\sqrt{1-v^2/c^2}(1+\sqrt{1-v^2/c^2})}$	-160.88	-0.74	DETECTED_UNRESOLVED ^a
Screened Coulomb ($r \leq 4.0$)	$\theta_2(1 - e^{-r/\theta_0})$	-1617.66	25.74	DETECTED_UNRESOLVED ^b
Entropy Expansion ($dV/V_i \leq 3.0$)	$\theta_3 \ln(1 + \frac{dV}{V_i})$	-811.34	37.99	IDENTIFIABLE ^c

^a Failed Gate V.2 (positive control) and Gate V.3 (ablation margin $\Delta\text{BIC} = -0.74 < 10.0$).

^b Passed Gate V.3 ($\Delta\text{BIC} = 25.74$), but failed Gate V.2 (positive control at 1% noise level).

^c Passed all four validation gates (Gate V.1–V.4).

TABLE VII. Aggregate epistemic tier distribution across all 165 evaluation instances (60 zero-noise runs + 105 noise-sweep runs). This explicitly demonstrates the separation between structural search capability and formal epistemic certification.

Regime	IDENTIFIABLE	DETECTED UNRESOLVED	WITHHELD
Zero Noise ($n = 60$)	60	0	0
Noisy ($n = 105$)	52	53	0
Total ($n = 165$)	112	53	0

widths of ≈ 16 –40 percentage points; larger-scale replication ($n \geq 50$) will further tighten confidence bounds.

- Epistemic qualification in high noise:** In the noise-sweep regime (1%–30%), 53 out of 105 runs yielded DETECTED_UNRESOLVED verdicts due to noise-induced ambiguity between competing functional forms, demonstrating that while ADCD avoids false-positive claims, high measurement noise naturally bounds formal structural certifiability.
- Baseline comparison framing:** Our comparison with PySR evaluates structural recovery under ADCD’s criteria (functional class recovery with extrapolation stability); PySR was evaluated under its default parsimony objective without dimensional penalty extensions, reflecting its role as an unconstrained exploratory baseline rather than a correction-specialized engine.
- Extended scenario suite and observational connectors:** In addition to the three canonical benchmark scenarios evaluated in this paper, the open-source repository implements an extended suite of 21 multi-variable coupled scenarios (including Yukawa forces, nonlinear damping, and blind held-out identity batteries) as well as preliminary observational connectors for historical experimental datasets (Mercury perihelion precession, SPARC galactic rotation curves, and Pantheon+ supernovae). While systematic multi-seed benchmarking on this extended suite represents ongoing work, the open-source infrastructure is released to facilitate community expansion.

VI. CONCLUSION

We presented ADCD, a correction-first symbolic regression framework with formal identifiability guarantees. Its core contributions are: algebraically regularized primitives that enforce classical-limit consistency by construction; a bounded grammar yielding strictly reproducible search via deterministic hashing; and a four-step epistemic validation protocol that explicitly reports when structural claims are and are not supported by the data.

Empirical evaluation across 165 unique data instances reveals that structural identifiability and numerical fit quality are independent dimensions of evaluation, a finding with implications beyond ADCD’s specific implementation. At zero noise, ADCD recovers the generating functional form in 90–100% of runs across all three benchmark scenarios, while an unconstrained baseline recovers it in 10–50%. By explicitly formalizing the conditions under which structural claims can and cannot be certified, ADCD provides a foundation for trustworthy, interpretable physical law discovery.

CODE AND DATA AVAILABILITY

The full implementation of ADCD, benchmark scenarios, evaluation scripts, and replication data are publicly available at <https://github.com/apiprdt/PhysicsPaper/tree/JuliaADCD> (branch: JuliaADCD). Full audit logs, raw run outputs ($n = 165$), and visualization scripts are included.

a. Implementation Architecture. ADCD was originally prototyped as a pure-Python/JAX implementation; the search and continuous optimization core has since been migrated to a native Julia (`Optim.jl`) production engine for high-throughput AST evaluation and robust parameter optimization (Section III D). The legacy Python/JAX path remains in the repository (selectable via `--engine python`) for cross-validation during migration but was not used to generate any of the results reported in this paper; all reported numbers use the au-

thoritative Julia backend (`--engine julia`, the default). Where configuration defaults between the two backends

differ (e.g., n_{restarts} , EBIC thresholds), the Julia values documented in Section III D are authoritative.

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Appendix A: Interfaces and Reproducibility

1. Input and Output Contracts

ADCD requires four mandatory inputs and accepts three optional inputs:

- `y_obs` (`np.ndarray`): Observed measurement array.
- `y_cl` (`np.ndarray`): Classical baseline prediction array.
- `X` (`dict[str, np.ndarray]`): Named feature arrays, e.g., `{"v": v_array}`.
- `units` (`dict[str, str]`): SI unit string per variable, e.g., `{"v": "m/s"}`.
- `domain` (`dict[str, float]`, optional): Observation window bounds.
- `taxonomy` (`str` | `None`, optional): Domain hint. If `None`, ADCD runs in *blind mode*.
- `delta_bic_threshold` (`float`, default = 10): Kass–Raftery threshold.

ADCD returns an `ADCDResult` dataclass containing the epistemic verdict (`tier`: "IDENTIFIABLE", "DETECTED_UNRESOLVED", or "WITHHELD"), `top_expression` (SymPy string), `top_latex`, `discovered_class`, `nmse_train`, `bic_score`, `pareto_front` (all surviving candidates ranked by BIC), and `checks` (complete machine-readable audit trail). `WITHHELD` does not mean “no output”; the full Pareto front and diagnostic summary are always accessible.

2. Minimal Python Usage and Primitive Registration

Listing VIII provides a self-contained Python example demonstrating both standard discovery and custom primitive registration.

Appendix B: Algorithm Details

1. Algorithm 2: Dynamic Correction Mode Detection

Algorithm B 1 formalizes the mode detection step referenced in Section III. The key design decision is the use of Spearman rank correlation rather than Pearson correlation, which provides robustness to non-linear monotone relationships and outliers.

2. Buckingham- π Dimensionless Argument Generation

The dimensionless argument set $\{u_{ij}\}$ is generated by exhaustive permutation of ratios and products of the input variables and classical constants, restricted to those that are dimensionless under the declared SI unit system $\mathcal{U}$. For example, given $\mathbf{X} = \{v\}$ with $\mathcal{U} = \{v : \text{m/s}\}$ and classical constant $c = 3 \times 10^8 \text{ m/s}$, the generator produces $u_1 = v/c$ and $u_2 = v^2/c^2$ (both dimensionless), while $v \cdot c$ (units m^2/s^2) is rejected. To aggressively constrain the search space, the generator implements strict heuristic pruning: polynomial degrees are capped (max degree = 2), and pure constant ratios (e.g., c_1/c_2) are discarded. Scale parameters θ_s with the same units as an input variable are automatically included in the permutation pool, allowing the algorithm to discover characteristic length, time, or energy scales (e.g., the Debye length $\lambda_D = \theta_0$ in the Screened Coulomb scenario).

Appendix C: Extended Benchmark Results

1. Noise-Level Sweep with 95% Confidence Intervals

Table IX extends Table V with Wilson 95% confidence intervals for each cell ($n = 5$ seeds per cell). The intervals confirm that at high noise (≥ 0.20), the confidence bands of ADCD and PySR do not overlap for any scenario, consistent with the structural claim that ADCD’s advantage is not an artifact of small-sample variation.

TABLE VIII. Minimal runnable Python example: (Left) standard ADCD discovery; (Right) custom primitive registration.

Standard ADCD Discovery Workflow

```

import numpy as np
from adcd.api import discover_correction

# 1. Define data (Time Dilation example)
v = np.linspace(0.01, 0.3, 200) # v in units of c
y_cl = 0.5 * v**2                # Classical KE
y_obs = y_cl * (1 + 0.01 * np.random.randn(200))

# 2. Run ADCD (Blind Search)
result = discover_correction(
    y_obs=y_obs, y_cl=y_cl,
    X={"v": v}, units={"v": "m/s"},
    domain={"v": 0.3}, taxonomy=None,
)

# 3. Read epistemic results
result.summary()                # prints full audit
print(result.tier)              # e.g., IDENTIFIABLE
print(result.top_expression)    # discovered formula
print(result.bic_score)        # BIC of Rank-1
chk = result.checks["ablation_control"]
print(f"Delta-BIC: {chk['bic_diff']:.2f}")

```

Registering Custom Primitives

```

from adcd.api import register_primitive

# Domain-specific Yukawa screened exponential:
# Satisfies asymptotic limit:  $\lim_{u \rightarrow 0} D(u) = 0$ 
register_primitive(
    name="D_yukawa",
    template="exp(-{u}) * {u}",
    domain_note="Yukawa decay with linear onset",
)

# Once registered, D_yukawa is automatically
# included in the grammar proposal pool:
result = discover_correction(
    y_obs=y_obs, y_cl=y_cl,
    X={"r": r}, units={"r": "m"},
)

```

Algorithm 2: *Dynamic Correction Mode Detection*

Input: Observed array y_{obs} , classical baseline y_{cl} , minimum sample size $n_{\text{min}} = 5$.
Output: Selected mode $m \in \{\text{additive}, \text{multiplicative}\}$, confidence $\kappa \in [0.5, 1.0]$.

```

1: function DETECTMODE( $y_{\text{obs}}, y_{\text{cl}}, n_{\text{min}}$ )
2:   if  $|\{y_i\}| < n_{\text{min}}$  or  $\text{Var}(y_{\text{cl}}) < 10^{-12}$  then
3:     return  $\langle \text{additive}, 0.5 \rangle$   $\triangleright$  degenerate fallback
4:   end if
5:    $r_{\text{add}} \leftarrow y_{\text{obs}} - y_{\text{cl}}$ 
6:    $r_{\text{mul}} \leftarrow y_{\text{obs}}/y_{\text{cl}} - 1$  (where  $|y_{\text{cl}}| > 10^{-12}$ )
7:   function SCALEDEPENDENCE( $y_{\text{cl}}, r$ )
8:      $\rho_{\text{cl}} \leftarrow \text{rank}(|y_{\text{cl}}|)$ ;  $\rho_r \leftarrow \text{rank}(|r|)$ 
9:     return  $|\text{Corr}(\rho_{\text{cl}}, \rho_r)|$   $\triangleright$  Spearman  $|\rho|$ 
10:  end function
11:   $d_{\text{add}} \leftarrow \text{SCALEDEPENDENCE}(y_{\text{cl}}, r_{\text{add}})$ 
12:   $d_{\text{mul}} \leftarrow \text{SCALEDEPENDENCE}(y_{\text{cl}}[m], r_{\text{mul}}[m])$ 
13:  if  $d_{\text{add}} + d_{\text{mul}} < 10^{-9}$  then
14:    return  $\langle \text{additive}, 0.5 \rangle$ 
15:  end if
16:  if  $d_{\text{add}} \leq d_{\text{mul}}$  then
17:     $\kappa \leftarrow 0.5 + 0.5 \cdot \frac{d_{\text{mul}} - d_{\text{add}}}{d_{\text{add}} + d_{\text{mul}}}$ 
18:    return  $\langle \text{additive}, \kappa \rangle$ 
19:  else
20:     $\kappa \leftarrow 0.5 + 0.5 \cdot \frac{d_{\text{add}} - d_{\text{mul}}}{d_{\text{add}} + d_{\text{mul}}}$ 
21:    return  $\langle \text{multiplicative}, \kappa \rangle$ 
22:  end if
23: end function

```

TABLE IX. Structural recovery rate (%) with Wilson 95% CI ($n = 5$ seeds per cell). Parenthetical ranges are $[\ell, u]$ Wilson intervals rounded to the nearest integer.

Noise (σ)	Time Dilation		Screened Coulomb		Entropy Expansion	
	ADCD	PySR	ADCD	PySR	ADCD	PySR
0.01	100 [57–100]	60 [23–88]	80 [37–96]	0 [0–43]	100 [57–100]	20 [4–62]
0.02	100 [57–100]	60 [23–88]	80 [37–96]	60 [23–88]	100 [57–100]	0 [0–43]
0.05	100 [57–100]	60 [23–88]	100 [57–100]	40 [12–77]	100 [57–100]	0 [0–43]
0.10	100 [57–100]	40 [12–77]	80 [37–96]	20 [4–62]	80 [37–96]	0 [0–43]
0.15	100 [57–100]	20 [4–62]	80 [37–96]	60 [23–88]	80 [37–96]	0 [0–43]
0.20	100 [57–100]	20 [4–62]	60 [23–88]	0 [0–43]	60 [23–88]	0 [0–43]
0.30	100 [57–100]	0 [0–43]	20 [4–62]	0 [0–43]	40 [12–77]	0 [0–43]